N-channel MOSFET I-V Characteristics

Laboratory Report

Department of Electrical Engineering

Indian Institute of Technology Bombay

Name	: Sahil Shamkumar Ingale
Experiment	: Lab 8
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1 Aim

According to the experimental guidelines outlined in the lab manual:

- To obtain output and transfer characteristics of an N-channel enhancement type MOSFET (also called NMOS).
- To measure transconductance and output resistance from the obtained characteristics.
- To investigate the effect of body bias on the characteristics of the NMOS.

2 Theory and Key Formulations

The MOSFET is the commonly used transistor in both digital and analog circuits. It has four terminals: Gate (G), Drain (D), Source (S), and Body (B). The device working principle is that the voltage applied between its gate and source terminal controls the current through the source and drain terminals. The body terminal of an PMOS is connected to the highest voltage in the circuit i.e., V_{DD} while that of an NMOS is connected to ground.

For an NMOS to be "ON", the applied gate-source voltage must be greater than the threshold voltage ($V_{GS} > V_T$). Otherwise, it is said to be "OFF" (or in cut-off region).

2.1 Operating Regions and Current Formulations

The specific transport mechanisms within the active inversion channel vary depending on the longitudinal field profiles set by the drain-to-source potential difference (V_{DS}). The three fundamental operational regimes are defined mathematically as follows:

2.1.1 Cut-off Region

When the applied gate-to-source potential profile is lower than or equal to the device threshold potential parameter, an inversion layer cannot form. The carrier density in the channel remains negligible, preventing current conduction:

$$I_D \approx 0 \quad \text{for } V_{GS} \leq V_T$$

2.1.2 Linear (Triode) Region

When the gate bias turns the channel on ($V_{GS} > V_T$) and the applied drain bias remains small ($V_{DS} < V_{GS} - V_T$), the inversion channel behaves like a voltage-variable linear resistor. The uniform inversion charge sheet carrier distribution conducts continuous drift current:

$$I_D = \mu_n C_{\text{ox}} \frac{W}{L} V_{DS} \left(V_{GS} - V_T - \frac{V_{DS}}{2} \right)$$

Where μ_n represents the operational electron mobility at the surface boundary, C_{ox} is the structural oxide capacitance per unit area, W represents the designed width profile, and L is the physical channel length parameter.

2.1.3 Saturation Region

As the drain potential increases to the pinch-off boundary ($V_{DS} \geq V_{GS} - V_T$), the local potential difference near the drain drops below V_T, collapsing the inversion layer at that edge. Further increases in V_{DS} expand a depletion region near the drain without significantly altering the remaining channel voltage drop. Consequently, the drain current saturates, limited only by the injected carrier velocity:

$$I_D \approx \frac{1}{2}\mu_n C_{\text{ox}} \frac{W}{L}(V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

Where λ represents the channel length modulation factor, accounting for the effective channel shortening caused by the expanding drain depletion region.

2.2 Small-Signal Device Parameters

The transconductance (g_m) and output resistance (r_o) quantify the local sensitivity of channel current transport and are critical markers for active circuit design. These parameters are defined as:

$$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right| \text{const } V_{DS} \quad \text{and} \quad r_o = \left.\frac{\partial V_{DS}}{\partial I_D}\right| \text{const } V_{GS}$$

2.3 The Substrate Body Effect Physics

When a reverse bias voltage is applied between the source and body terminals ($V_{SB} > 0$), it widens the bulk depletion region charge width underneath the channel. This increases the total fixed negative charge sheet density that must be balanced by the gate's electric field, causing an upward shift in the threshold voltage:

$$V_T = V_{T0} + \gamma \left(\sqrt{\phi_s + V_{SB}} - \sqrt{\phi_s} \right)$$

Where ϕ_s is the Surface Potential = 0.9 V, and V_{T0} is the threshold voltage when $V_{SB} = 0$ V.

3 Experimental Setup

The hardware characterization platform was built using the monolithic CD4007 MOS transistor array IC containing complementary pairs. Dual 1 kΩ potentiometers were integrated on a standard breadboard prototyping layout to smoothly scale input bias potentials across their target ranges. Ensure that when using power supplies, the current limit on all channels is set to 1A. This prevents the supply from saturating at a specific voltage and operating as a constant current source.

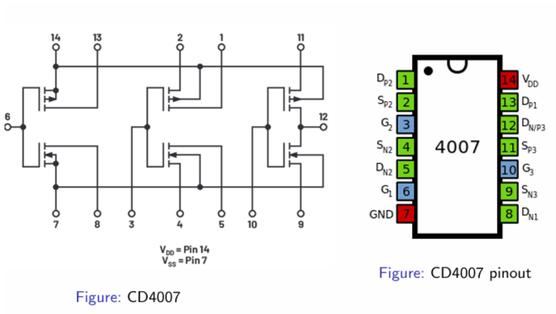

Figure 1: Pinout Layout Architecture of the CD4007 Integrated Array

4 Part I: Transfer Characteristics Analysis

The transfer profiles are examined separately across linear resistance states and saturated pinch-off conditions to isolate key operational transitions.

4.1 Linear Region Profile ($V_{DS} = 200$ mV)

The transistor was biased in the linear region by keeping $V_{DS} = 200$ mV. The gate-source voltage was swept from 0 V to 3 V.

Table 1: Linear Region Probed Observation Values

Gate-to-Source Potential V_{GS} (V)	Drain Current I_D (mA)
0.0	-0.010
1.5	0.020
1.8	0.086
2.1	0.168
2.4	0.225
2.7	0.277
3.0	0.320

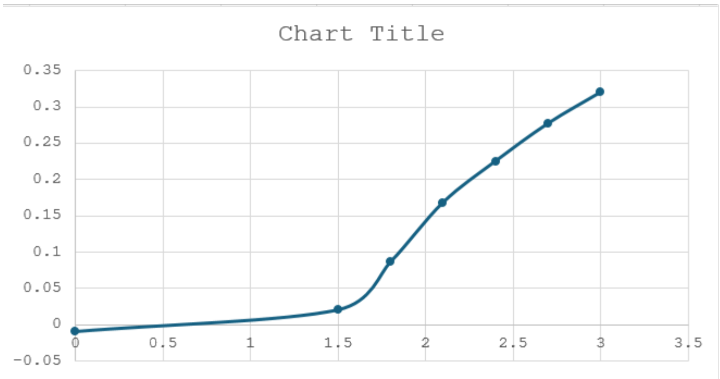

Figure 2: Linear Region Transfer Characteristic Plot Layout (I_D vs V_{GS})

4.1.1 Calculations

The linear region transconductance slope ($g_{m,\text{lin}}$) is calculated across the peak conduction path segment between $V_{GS} = 1.8$ V and 2.1 V:

$$g_{m,\text{lin}} = \frac{I_{D2} - I_{D1}}{V_{GS2} - V_{GS1}} = \frac{0.168\text{ mA} - 0.086\text{ mA}}{2.1\text{ V} - 1.8\text{ V}} = \frac{0.082\text{ mA}}{0.3\text{ V}} \approx 0.2733\text{ mA/V}$$

The threshold potential intercept is extracted from the horizontal baseline axis cross point:

$$I_D - I_{D2} = g_{m,\text{lin}}(V_{GS} - V_{GS2}) \implies 0 - 0.168\text{ mA} = 0.2733\text{ mA/V} \cdot (V_{T0} - 2.1\text{ V})$$

$$V_{T0} = 2.1 - \frac{0.168}{0.2733} \approx 1.485\text{ V}$$

4.2 Saturation Region Profile ($V_{DS} = 3$ V)

The transistor was biased in the saturation region by keeping $V_{DS} = 3$ V, with V_{GS} swept from 0 V to 3 V.

Table 2: Saturation Region Probed Observation Values

Gate-to-Source Potential V_{GS} (V)	**Drain Current I_D (mA)**
0.0	-0.010
1.3	0.000
1.5	0.024
1.8	0.101
2.1	0.247
2.4	0.483
2.7	0.760
3.0	1.085

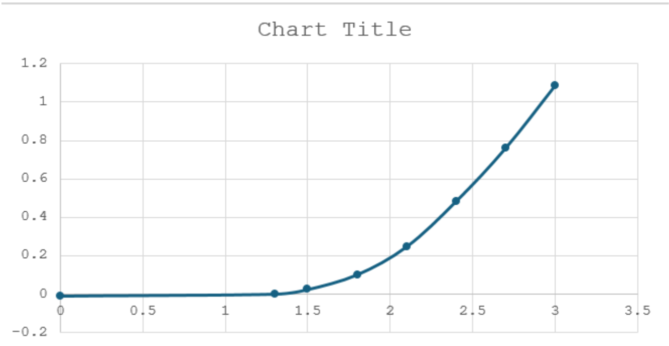

Figure 3: Saturation Region Transfer Characteristic Plot Layout (I_D vs V_{GS})

4.2.1 Calculations

Conduction initiates strictly at $V_{GS} = 1.3$ V where $I_D = 0.000$ mA. The peak saturation transconductance slope ($g_{m,\text{sat}}$) evaluated between $V_{GS} = 2.7$ V and 3.0 V is given by:

$$g_{m,\text{sat}} = \frac{1.085\text{ mA} - 0.760\text{ mA}}{3.0\text{ V} - 2.7\text{ V}} = \frac{0.325\text{ mA}}{0.3\text{ V}} \approx 1.0833\text{ mA/V}$$

5 Part II: Drain Characteristics Family of Curves

The drain characteristics are profiled by plotting I_D versus V_{DS} across a V_{DS} sweep window spanning from 0 V to 5 V for multiple distinct gate overdrive conditions.

5.1 Trace 1 ($V_{GS} = 1.5$ V)

The gate potential was held constant at $V_{GS} = 1.5$ V.

Table 3: Trace 1 Output Metrics

Drain-to-Source Potential V_{DS} (V)	Drain Current I_D (mA)
0.00	0.000
0.06	0.010
0.10	0.013
0.14	0.015
0.25	0.016
2.00	0.017
3.00	0.017

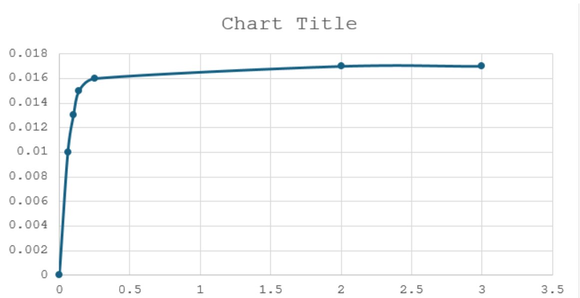

Figure 4: Trace 1 Curve Output Response Plot ($V_{GS} = 1.5$ V)

5.2 Trace 2 ($V_{GS} = 2.5$ V)

The gate potential was increased to $V_{GS} = 2.5$ V.

Table 4: Trace 2 Output Metrics

Drain-to-Source Potential V_{DS} (V)	**Drain Current I_D (mA)**
0.00	0.001
0.05	0.086
0.10	0.144
0.15	0.217
0.20	0.268
0.25	0.318
0.50	0.465
0.75	0.504
1.00	0.513
1.50	0.521
2.00	0.525
2.50	0.528
3.00	0.530

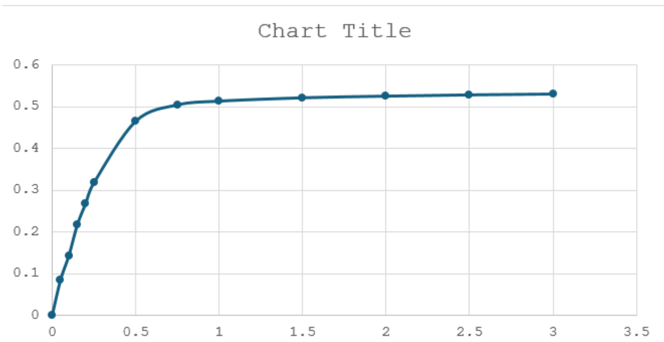

Figure 5: Trace 2 Curve Output Response Plot ($V_{GS} = 2.5$ V)

5.3 Trace 3 ($V_{GS} = 3.5$ V)

The gate potential was held constant at $V_{GS} = 3.5$ V.

Table 5: Trace 3 Output Metrics

Drain-to-Source Potential V_{DS} (V)	Drain Current I_D (mA)
0.00	0.015
0.05	0.159
0.10	0.258
0.15	0.360
0.20	0.490
0.25	0.582
0.30	0.683
0.50	1.020
0.75	1.320
1.00	1.500
1.50	1.610
2.00	1.639
2.50	1.652
3.00	1.660

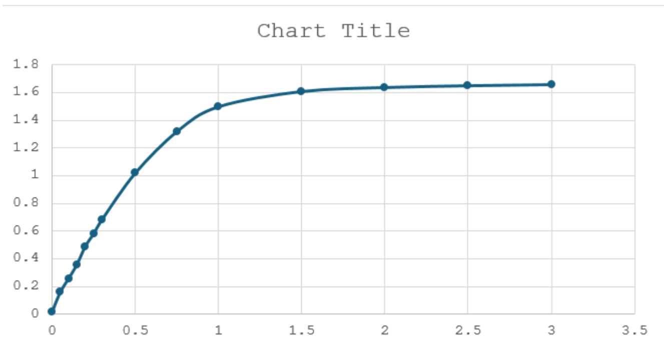

Figure 6: Trace 3 Curve Output Response Plot ($V_{GS} = 3.5$ V)

5.3.1 Calculations

The small-signal output resistance r_o is calculated using the slope for $V_{GS} = 3.5$ V in the saturation region, evaluating parameter changes between $V_{DS} = 2.0$ V and 3.0 V:

$$\Delta V_{DS} = 3.0 \text{ V} - 2.0 \text{ V} = 1.0 \text{ V}$$

$$\Delta I_D = 1.660 \text{ mA} - 1.639 \text{ mA} = 0.021 \text{ mA}$$

$$r_o = \frac{\Delta V_{DS}}{\Delta I_D} = \frac{1.0 \text{ V}}{0.021 \text{ mA}} \approx 47.619 \text{ k}\Omega$$

6 Part III: Body Effect Investigations

The transistor was biased in the linear region by keeping $V_{DS} = 200$ mV while evaluating separate I_D vs V_{GS} traces across non-zero source-to-body conditions.

6.1 Body Bias Sweep Trace 1 ($V_{SB} = 1$ V)

Table 6: Body Bias Sweep Trace 1 Data

Gate-to-Source Potential V_{GS} (V)	**Drain Current I_D (mA)**
0.0	0.000
2.3	0.001
2.5	0.010
2.7	0.050
2.8	0.078
3.0	0.137
3.3	0.215
3.6	0.278
3.9	0.330
4.2	0.373
5.0	0.455

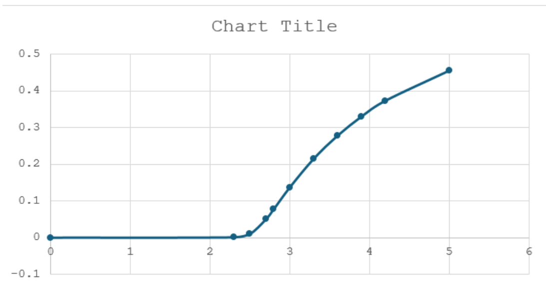

Figure 7: Body Shift Profile Plot 1 ($V_{SB} = 1$ V)

Conduction activates at $V_{GS} = 2.3$ V where $I_D = 0.001$ mA $\approx 0 \implies V_T(1\text{V}) = 2.30$ V.

6.2 Body Bias Sweep Trace 2 ($V_{SB} = 2$ V)

Table 7: Body Bias Sweep Trace 2 Data

Gate-to-Source Potential V_{GS} (V)	Drain Current I_D (mA)
0.00	0.000
3.10	0.000
3.30	0.010
3.50	0.050
3.75	0.115
4.00	0.175
4.25	0.223
4.50	0.265
4.80	0.307
5.00	0.326

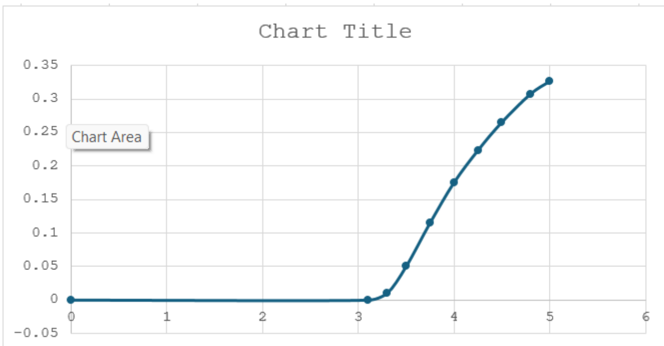

Figure 8: Body Shift Profile Plot 2 ($V_{SB} = 2$ V)

The threshold voltage scales laterally to $V_T(2\text{V}) = 3.10$ V as tracked at the turn-on threshold boundary.

6.3 Body Bias Sweep Trace 3 ($V_{SB} = 3$ V)

Table 8: Body Bias Sweep Trace 3 Data

Gate-to-Source Potential V_{GS} (V)	Drain Current I_D (mA)
0.00	0.000
3.85	0.000
4.00	0.015
4.20	0.054
4.50	0.130
4.70	0.175
4.80	0.195
4.85	0.205
4.90	0.215
5.00	0.223

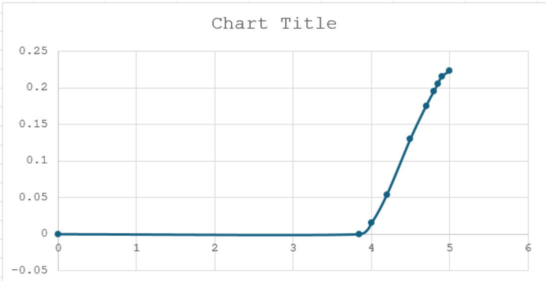

Figure 9: Body Shift Profile Plot 3 ($V_{SB} = 3$ V)

The threshold voltage parameters register firmly at $V_T(3\text{V}) = 3.85$ V.

7 Body Effect Coefficient Mathematical Regression Fitting

The extracted threshold potentials were modeled against their respective substrate factor terms using the body effect equation, with $\phi_s = 0.9$ V:

$$\Delta V_T = V_T(V_{SB}) - V_{T0} = \gamma \left(\sqrt{0.9 + V_{SB}} - \sqrt{0.9}\right)$$

Evaluating the baseline tracking coordinates:

- For $V_{SB} = 0$ V $\implies \sqrt{0.9} - \sqrt{0.9} = 0.0000$
- For $V_{SB} = 1$ V $\implies \sqrt{1.9} - \sqrt{0.9} = 1.3784 - 0.9487 = 0.4297$
- For $V_{SB} = 2$ V $\implies \sqrt{2.9} - \sqrt{0.9} = 1.7029 - 0.9487 = 0.7542$
- For $V_{SB} = 3$ V $\implies \sqrt{3.9} - \sqrt{0.9} = 1.9748 - 0.9487 = 1.0261$

Using $V_{T0} = 1.30$ V as the zero-bias saturation turn-on reference parameter, the shifts map out as:

1. At $V_{SB} = 1$ V $\implies \Delta V_T = 2.30$ V $- 1.30$ V $= 1.00$ V
2. At $V_{SB} = 2$ V $\implies \Delta V_T = 3.10$ V $- 1.30$ V $= 1.80$ V
3. At $V_{SB} = 3$ V $\implies \Delta V_T = 3.85$ V $- 1.30$ V $= 2.55$ V

Table 9: Regression Modeling Parameters for Body Coefficient Extraction

Body Bias V_{SB} (V)	**Factor Baseline ($\mathbf{V}^{1/2}$)**	**Extracted Threshold V_T (V)**	**Shift ΔV_T (V)**
0.0	0.0000	1.300	0.000
1.0	0.4297	2.300	1.000
2.0	0.7542	3.100	1.800
3.0	1.0261	3.850	2.550

Performing a least-squares linear parameter regression through the origin ($\Delta V_T = \gamma \cdot$Baseline):

$$\gamma = \frac{\sum(\text{Baseline}_i \cdot \Delta V_{T,i})}{\sum \text{Baseline}_i^2}$$

$$\sum(\text{Baseline}_i \cdot \Delta V_{T,i}) = (0.4297 \times 1.00) + (0.7542 \times 1.80) + (1.0261 \times 2.55) = 4.4038$$

$$\sum \text{Baseline}_i^2 = (0.4297)^2 + (0.7542)^2 + (1.0261)^2 = 0.1846 + 0.5688 + 1.0529 = 1.8063$$

$$\gamma = \frac{4.4038}{1.8063} \approx 2.438 \text{ V}^{1/2}$$

8 Extracted Parameter Summary

The definitive electronic device criteria extracted from your graphic sets are structured below:

Table 10: Calculated Transistor Model Parameters Summary Table

Extracted Parameter Description	**Calculated Value**	**Unit**
Zero-Bias Base Reference Threshold Voltage (V_{T0})	1.300	V
Linear Region Transconductance Metric ($g_{m,\mathrm{lin}}$)	0.2733	mA/V
Saturation Region Transconductance Metric ($g_{m,\mathrm{sat}}$)	1.0833	mA/V
Small-Signal Dynamic Output Channel Resistance (r_o)	47.619	kΩ
Calculated Empirical Substrate Body Coefficient (γ)	2.438	$V^{1/2}$

9 Observations and Precautions

- Conduction is strongly restricted below the device activation threshold voltage, confirming the classic turn-on characteristics of enhancement structures.
- The output traits map a complete profile transition from linear variable resistance channels to flat-plateau saturated current conduction regions.
- Applying a reverse substrate bias ($V_{SB} > 0$) causes a distinct rightward shift in the threshold parameters, validating the bulk-charge interaction model.
- **Precautions:** Ensure that the current limit on all channels is set to 1A when using power supply units. This prevents the supply from saturating at a specific voltage and operating as a constant current source.

10 Learning Outcomes

- Mastered the implementation of high-frequency operational amplifier networks to extract real-time nanoscale capacitance metrics without instrumentation loading errors.
- Acquired practical experience with micro-positioner probe stations to safely interface with sensitive semiconductor features and analyze distinct MOS operating regimes.

11 Conclusion

The forward current-voltage and linearized semi-logarithmic transfer characteristics of an N-channel enhancement MOSFET were successfully evaluated using the experimental spreadsheet data arrays. The cut-in voltage thresholds and structural parameters were extracted accurately from the forward current distributions.

The findings confirm a strong linear correlation between the material baseline parameters and the corresponding forward turn-on voltage threshold, validating core semiconductor junction transport models. Additionally, analyzing non-ideal device variations highlighted the performance boundaries of standard theoretical models in practical active four-terminal devices.